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SMART DATA INSIGHT AND PATTERN ANALYZER: AN AI-BASED APPROACH TO INTELLIGENT DATA ANALYTICS

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Abstract:

The increasing volume of digital data creates a need for efficient and intelligent data analysis. Traditional methods often require manual processing, making pattern identification and decision-making time-consuming. To address this challenge, this project proposes the Smart Data Insight and Pattern Analyzer, an AI-based platform that automates data processing, pattern recognition, and visualization.

The system uses Python and Flask for backend development, with HTML, CSS, and JavaScript for the user interface. It supports CSV, JSON, and Excel datasets, utilizing Pandas and NumPy for preprocessing and machine learning techniques for classification and anomaly detection. The platform also provides interactive visualizations, confidence scores, and feature importance summaries to simplify data interpretation.

By integrating data processing, machine learning, and visualization into a single platform, the proposed system reduces manual effort and improves analytical efficiency. It provides an accessible solution for identifying meaningful insights and supporting data-driven decision-making.

Keywords: Data Analytics, Artificial Intelligence, Machine Learning, Pattern Recognition, Anomaly Detection, Data Visualization, Flask, Predictive Analytics.